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Paper Title: Exploring the Nature and Variants of Relevance Feedback

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Abstract:

In this paper we analyze the nature of the relevance feedback problem in a continuous representation space in the context of multimedia information retrieval. Emphasis is put on exploring the uniqueness of the problem and comparing the assumptions and merits of various solutions in the literature. An attempt is made to compile a list of critical issues to consider when designing a relevance feedback algorithm. With a state-of-the-art review as the main portion, this paper also presents some novel solutions and perspectives throughout the discussion.

Keywords: relevance feedback, content-based image retrieval

Exploring the Nature and Variants of Relevance Feedback

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Abstract

In this paper we analyze the nature of the relevance feedback problem in a continuous representation space in the context of multimedia information retrieval. Emphasis is put on exploring the uniqueness of the problem and comparing the assumptions and merits of various solutions in the literature. An attempt is made to compile a list of critical issues to consider when designing a relevance feedback algorithm. With a state-of-the-art review as the main portion, this paper also presents some novel solutions and perspectives throughout the discussion.

1. Introduction

Initially developed in document retrieval[24], relevance feedback was transformed and introduced into content-based multimedia retrieval, mainly content-based image retrieval (CBIR), during mid 1990's[19][22]. Interestingly, it appears to have attracted more attention in the new field than the previous—a variety of solutions has been proposed within a short period of time and it remains an active research topic. The reasons can be that *more ambiguities arise when interpreting images than words*, which makes user interaction more of a necessity; and in addition, *judging a document takes time while an image reveals its content almost instantly to a human observer*, which makes the feedback process faster and more sensible for the end user.

The need for user-in-the-loop in CBIR stems from the fact that images reside in a continuous representation space, in which semantic concepts are best described in discriminative subspaces—"cars" are of certain *shape* while "sunset" is more describable by *color*. More importantly, different users at different times may have different interpretations or requests regarding the same image, which makes off-line learning unfeasible in general. A straightforward solution is to ask the user to tune the system parameters during retrieval process, but it is too much a burden for a common user[22]. A more feasible form of interaction is to ask the user to provide feedbacks regarding the (ir)relevance of the current retrieval results. The system then learns from these training examples to achieve an improved performance next round, iteratively if the user so desires.

Relevance feedback algorithms have been shown to provide dramatic performance boost in retrieval systems ([14][19][22][30][32] [33][35], etc).

2. The Characteristics of the Relevance Feedback Problem

Since the general assumption is that every user's need is different and time-varying, the database cannot adopt a fixed clustering structure; and *the total number of classes* and *the class membership* are not available in advance since these are assumed to be user-dependent and time varying as well. (Of course these rather extreme assumptions can be relaxed in a real application to the degree of choice. More arguments are presented in Section 4.)

If each image/region is represented by a point in a feature space, relevance feedback with only positive (i.e., relevant) examples can be cast as a density estimation [11][15] or novelty detection [27][3] problem; while with both positive and negative training examples it becomes a classification problem, or an on-line learning problem in a batch mode, but with the following unique characteristics:

Small sample issue. The number of training examples is small (typically <20 per round of interaction) relative to the dimension of the feature space (from dozens to hundreds, or even more), while the number of classes is large. For such small sample size, some existing learning machines such as support vector machines(SVM) [31] cannot give meaningful results [30][41].

Asymmetry in training sample. The desired output of information retrieval is not necessarily a binary decision on each point as given by a classifier, but rather a rank-ordered top-k returns. This is a less demanding task since the rank or configuration of the irrelevant classes/points is of no concern as long as they are well beyond the top-k returns. Most classification or learning algorithms, e.g., discriminant analysis [5] or SVM [31], treat the positive and negative examples interchangeably and assume that both sets represent the true distributions well. But in reality the small number of negative examples is unlikely to be representative for all the irrelevant classes; thus an asymmetric treatment may be necessary [41].

Real time requirement Finally, since the user is interacting with the machine in real time, the algorithm shall be sufficiently fast, and avoid if possible heavy computations over the whole dataset.

3. The Variants of Relevance Feedback Algorithms

It is not the intention of this section to review *all* existing techniques but rather to point out the major variants and compare their merits. We would emphasize that *under the same notion of "relevance feedback", different methods might have adopted different assumptions or problem settings thus incomparable*. The following lists some of the conceptual dimensions along which some methods greatly differ from others:

a) What is the user looking for?

While most of the work assume the user is looking for "a class of similar items" to the query at hand, Cox et al. [4] assume that the user is looking for "a particular target item" and that the feedback is in the form of "relative judgment", i.e., the positive items are not necessarily the target but closer to the target than others. They proposed the use of *stochastic comparison search* as the relevance feedback algorithm.

b) What to feedback, or, what is the training data?

Some algorithms assume the user will give a binary feedback for positive and negative examples [30]; some only take positive examples [11][23][3]; some take positive and negative examples with "degree of (ir)relevance" for each [22]; some assumes the feedback is only a "comparative judgment" instead of a definite hit or miss [4]; Some uses both labeled and unlabeled data for training: Wu et al. [38] proposed D-EM algorithm within the transductive learning framework and used examples from the user feedback (labeled data) as well as other data points (unlabeled data). It performs discriminant analysis inside the EM iterations to select a subspace of features, such that the two-class (positive and negative) assumption on the data distributions has better support. The results were promising, but the computation can be a concern for the EM iterations in real time for large dataset.

A novel form of training is "learning from layout of images" during browsing or data visualization process [25][29]. The idea is to ask the user to layout images on a "table" or to manipulate an existing layout (which can be obtained through multidimensional scaling techniques), according to the user's interpretation of the semantic relationships among the images. The machine will then learn a feature weighting or transformation from the relative image positions [29].

c) Feature selection and representation.

In terms of feature selection, while most CBIR systems use traditional image features such as color histogram or moments, texture, shape, and structure features, there are alternatives. Tieu and Viola [30] used more than 45,000 "highly selective features", and a boosting technique to learn a classification function in this feature space. The features were demonstrated to be sparse with high kurtosis, and were argued to be expressive for high-level semantic concepts. Weak 2-class classifiers were formulated based on Gaussian assumption for both the positive and negative (randomly chosen) examples along each feature component, independently. The strong classifier is a weighted sum of the weak classifiers as in AdaBoost [7]. As for feature representation, while most assume one feature vector per image/region as the basic representation, Vasconcelos and Lippman [32] adopted Gaussian mixture model on DCT coefficients as image representation. Bayesian inference is then applied for classification and learning over time. Richer information captured by the mixture model also makes image regional matching possible.

d) Class distribution.

Another issue is what assumption to be imposed on the target class(es). Gaussian assumption is a common and convenient choice [11][23]. Wu et al. [36] treated multiple queries as a disjunctive set and used an aggregate dissimilarity function to combine for a candidate image the pair-wise distances to every positive example as the distance measure. This shall be compared to a Parzen window method [15], in which Parzen window density estimation was applied to capture non-linearity in the distribution of positive examples. An elegant way to deal with non-linearity is to use reproducing kernel based algorithms. A kernel based one-class SVM as density estimator for positive examples was shown [3] to outperform the whitening transform based linear/ quadratic method. The density estimation based schemes usually lack discriminative power since they ignore negative feedbacks [36][3].

e) Data organization.

If a hierarchical tree structure is adopted in the database for more efficient access [2], the learning becomes more difficult since the tree-structure needs to be updated in real time. The trade-off offered by [2] between the speed and accuracy in searching becomes crucial. More discussions regarding this issue is put in Section 4.

f) The goals and the learning machines

Along this line lies the greatest variability among different methods. A number of the early methods—self-labeled as "relevance feedback"[17][18][22][25] or not [13]—propose to learn a new query and the relative importance of different features or feature components, while others learn a linear transformation in the feature space taking into account correlations among feature components [11][23][41]. Some of the latest work treats it either as a density estimation [36][3], learning [14][30], or classification [32][38] problem.

In its short history, relevance feedback developed along the path from heuristic based techniques to optimal learning algorithms, with early work inspired by term weighting and relevance feedback techniques in document retrieval [24]. These methods proposed heuristic formulation with empirical parameter adjustment, mainly along the line of independent axis weighting in the feature

space[19][22][18][20][25][1]. The intuition is to emphasize more on the feature(s) that best clusters the positive examples and separates the positive and the negative.

Early works [19][22] had clear birthmarks from document retrieval field. For example, in [22], learning based on "term frequency" and "inverse document frequency" in text domain was transformed into learning based on the ranks of the positive and negative images along each feature axis in the continuous feature space. [19] quantized the features and then grouped the images or regions into hierarchical trees whose nodes were constructed through single-link clustering. Then weighting on groupings was based on set operations.

Kohonen's *learning vector quantization* (LVQ) algorithm [34] and tree-structured *self-organizing map* (TS-SOM) [12] were used for dynamic data clustering during relevance feedback. Laaksonen et al. [12] used TS-SOMs to index the images along different feature axes such as color and texture. Positive and negative examples were mapped to positive and negative impulses on the maps and a low-pass operation on the maps was argued to *implicitly* reveal the relative importance of different features because a "good" map will keep positive examples cluster while negative examples scatter away. This was based on a similar intuition as that of [18], where a probabilistic method was used instead to capture feature relevance. The assumption of feature independence imposed in these methods is rather artificial.

Later on researchers began to look at this problem from a more systematic point of view by formulating it into an optimization, learning, or classification problem. In [11] and [23], based on the minimization of total distances of positive examples from the new query, the optimal solutions turned out to be the weighted average as the new query and a whitening transform (or Mahalanobis distance metric) in the feature space. Additionally, Rui and Huang [23] adopted a two-level weighting scheme to better cope with singularity issue due to the small number of training samples. To take into account the negative examples, Schettini et al. [26] updated the feature weights along each feature axis by comparing the variance of positive examples to the variance of the union of positive and negative examples.

MacArthur et al. [14] cast relevance feedback as a two class learning problem, and used a decision tree algorithm to sequentially "cut" the feature space until all points within a partition are of the same class. The database was classified by the resulting decision tree: images that fall into a relevant leaf were collected and the nearest neighbors of the query were returned.

Some of the following approaches are off-line learning algorithms but have the potential for on-line application. Guo and Zhang [9] used AdaBoost for face recognition and audio retrieval. In [9], a constrained majority voting (CMV) strategy is proposed to speed up the pair-wise comparisons for multi-class classification. Note that in their case labeled training samples are available for all classes.

There were also schemes for learning object structure from examples based on image segmentation. Xu, Saber, and Tekalp [39] proposed a hierarchical formation scheme for object description from elementary regions determined by a segmentation using color and edge. From examples, the system learns a "composite node" of several regions with an adjacency matrix representing their spatial relationships. Ratan et al. [21] used *multiple-instant learning* model to learn the most important sub-image(s) from example images, which are represented as a bag (collection) of instances (subimages). The adopted *Diverse Density algorithm* tries to find the area in feature space that are shared by all positive images while far from all negative subimages. Along the same line is the work by Forsyth and Fleck [6], where the system learns a "body plan" for object. The difficulty in such representations is to achieve the invariance to viewing angle changes.

4. Issues to Consider When Designing a Relevance Feedback Algorithm

a) Negative examples

How to treat the *small number* of negative examples may be the center issue when negative feedbacks are to be considered. Tieu and Viola [30] used random sampling to get around the small sample issue, taking a risk of treating positive points as negative training samples. Vasconcelos and Lippman[33] assumed that a negative example for class S_i shall be a positive example for the complement of class S_i, and quantified in terms of likelihoods as follows:

$$P(\bar{y} \mid S_i = 1) = P(y \mid S_i = 0) \qquad (1)$$

where $\bar{y}$ means that y is treated as a negative example. Special steps are needed to avoid over-emphasizing the importance of negative examples.

Zhou and Huang [41] used the intuition that "all positive examples are alike in a way, each negative example is negative in its own way", and proposed asymmetric treatment for the positive and negative examples. They assumed that the positive examples have a more compact support while used the negative examples only to provide local discriminative information.

b) Singularity issue in sample covariance matrix.

Many relevance feedback algorithms used the sample covariance matrix and its inverse [11][22][23][26][41]. When the number of training examples is smaller than the dimensionality of the feature space, the singularity issue arises. The substitution of the *Moore-Penrose inverse* or *pseudo-inverse matrix* for the regular inverse proposed in [11] is not only mathematically unfounded, but also counterintuitive: Imagine a diagonal covariance matrix with the ith diagonal element being 0; according to the "weight by the inverse of the variance" heuristic implied in this method, this indicates that the ith axis of the feature

space is very expressive thus shall receive a very high weight. However, [11] will put a weight of zero on the ith axis.

Another proposal was to adopt a hierarchical weighting scheme (assuming a *block diagonal* matrix instead of the full covariance matrix) so that less parameters need to be estimated, and in the extreme case, just use a diagonal matrix[23]. *This implies a forced independence assumption*, which is too strong and can yield extremely biased eigenvalue estimations: as an example, assuming two positive examples $[0, 0]^T$ and $[1, 1]^T$, the sample covariance matrix C is

$$C = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}; \qquad C_{diag} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}. \tag{2}$$

Only estimating the diagonal matrix C_{diag} will result in equal weighting of the two axes by the weight vector $[2, 2]^T$. While an apparently better solution is to rotate the space clockwise by 45°, and then weight the vertical axis more. Additionally, invertibility is still not guaranteed even only diagonal elements are estimated.

It is known that under small sample the sample covariance matrix is statistically biased in the sense that the large eigenvalues become larger and the small smaller. A statistically valid solution is to add regularization terms on the diagonal of the sample covariance matrix before the inversion, which is also known as a "linear shrinkage estimator" [8][10][41]:

$$\hat{C} = (1-\mu)C + \frac{\mu}{n} tr[C] I \tag{3}$$

This operation, although simple and seemingly unintentional, actually compensates the aforementioned bias [8][10]. Following the example above:

$$\hat{C} = \begin{bmatrix} 0.5+0.01 & 0.5 \\ 0.5 & 0.5+0.01 \end{bmatrix}; \hat{C}^{-1} = V \Lambda V^{-1}; \\ V = \begin{bmatrix} -0.707 & -0.707 \\ -0.707 & 0.707 \end{bmatrix}; \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} \tag{4}$$

The solution in Equation(4) provides the rotation of 45° (by V) followed by a proper weighting of the axes (by Λ).

To this point it is worth noting that a unifying view of relevance feedback algorithms can be of "learning an optimal transform in the feature space" because: when only positive examples are considered, the "generalized ellipsoid distance metric"[11] is equivalent to a whitening transform followed by an Euclidean metric[23] since the eigenvalues are also the singular values; when negative examples are considered using discriminant analysis as in [41], the generalized eigenvalues are not the same as the singular values, and the solution is a "generalized whitening transform" followed by the Euclidean metric.

c) Feature normalization

Different image features need to be normalized to have comparable statistics, say normal distribution. Not surprisingly, this normalization can also be extended to a transformation of the feature space. From a discrimination point of view, the optimal normalization shall be the transform which separates *all* the semantically meaningful classes in the dataset while clusters within each class. Since the class membership is not known *a priori*, one possible solution is to use the accumulated feedbacks from all the users as the training set to be fed into a multiple discriminant analysis [5] framework to yield a transform that is optimal for the training data available so far. This implies more computation but can give better performance than the straightforward normal distribution assumption. A simplified example is that if all users emphasize *color feature* more than *texture* in *all* cases, there is no reason for maintaining equal variances along these two axes—stretching the *color* axis can give better initial retrieval result. This is illustrated in Figure 1. Note the "stretching direction" does not have to be aligned with the original axes as shown in the figure—correlations among axes can be modeled as well.

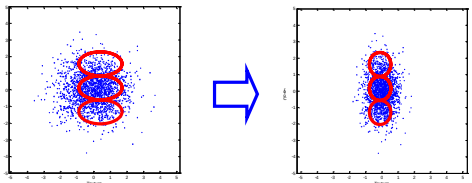


Figure 1 Feature normalization as a discriminating transformation: assuming all meaningful classes have ellipsoidal contours as shown.

d) Pre-clustering and relevance feedback

It may be argued that unsupervised clustering techniques—EM using minimum description length criteria, or mean shift—can determine the number of clusters in the database automatically. But semantically meaningful clustering depends on the subspace selection: an image of a "white elephant" in the feature space is not necessarily closer to a "gray elephant" than it is to a "white hippo", unless a proper discriminating subspace (e.g., *nose length*) can be specified beforehand—which is, however, exactly what relevance feedback is trying to learn in the first place. So in principle, *the rationale of relevance feedback contradicts that of pre-clustering*. However in reality, prior domain knowledge, if holds for all users of the system, can be used to guide a pre-organization of the dataset to facilitate faster retrieval. And a trade-off needs to be made between flexibility and efficiency. Chen, Bouman, and Dalton [2] provided a fast algorithm to update a hierarchical clustering structure during relevance feedback.

e) Global vs. regional query

Most relevance feedback schemes are designed to deal with global image features only. Some algorithms can be extended to deal with image blocks using a *concatenated* feature vector as the representation, and a hierarchical weighting scheme automatically reveals the relative importance of different image blocks during user interactions[40]. Vasconcelos and Lippman used Bayesian

inference [33] on image local features for relevance feedback learning. This scheme is inherently capable of image regional query, given that supervised training has been conducted on regional features to provide the priors.

f) Nearest neighbor search

When the size of the dataset is large and the dimensionality of the representation space is high, even a simple nearest neighbor search (under changing distance metric) can be computationally formidable for real time performance. One solution can be an adaptive nearest neighbor search [37], which updates the nearest neighbors intelligently and efficiently from one iteration to the next without searching the whole dataset repeatedly.

5. Conclusion

Relevance feedback during multimedia information retrieval is an on-line learning problem with distinct characteristics and difficulties. A successful algorithm is the one tailored to address these special issues.

As one of the future directions, we believe the integration of relevance feedback with multi-dimensional indexing techniques is worth further investigation for the application in large-scale multimedia databases.

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